

Airbnb Text Analytics: Business Value from Listing Descriptions

Valentine Dube - Data Visualization and Text Analytics Report

Executive Summary

This report examines the results from analyzing more than 5000 Airbnb listings through sentiment analysis with NRC, BING, AFINN lexicons, TF-IDF, Bigram, and LDA topic modeling techniques. The results show that positive sentiments can increase prices by 18% (\$143 vs. \$121), unique descriptors increase engagement by 30%, and themes based on locations increase premiums by 20-30%. Four R Shiny visualizations provide actionable insights on potential revenue increases of 30-40% through content optimization.

Text Mining Frameworks

Sentiment Analysis, AFINN uses numerical scores -5 to +5 aggregated at the listing level, BING uses binary classification, and NRC uses eight emotions. Results show that positive emotions dominate at 52%. Top-performing listings have +18 average sentiment compared to a +8.5 average in the overall dataset. Each 10-point improvement equates to \$15-20 price increases. Positive words like "comfortable," "beautiful," and "enjoy" dominating listings. Emotions are key value determinants.

TF-IDF and Bigrams: TF-IDF uses word uniqueness to weight words. Results show that 70% of listings use generic words like "nice" and "comfortable," which provide zero differentiation. Unique words like "villa," "loft," and "guesthouse" shows 30% higher engagement through review volume.

Topic Modeling: Latent Dirichlet Allocation k=6 topics. Results show that listings fall into urban living, beach living, apartment living, and three other topics. The topic distribution shows

relatively balanced market share of 15-18%. However, price analysis reveals that beach living topics have 20-30% premiums over generic urban apartments.

R shiny Dashboard: The dashboard is composed of various visualizations that show analytical insights along with KPIs such as total number of listings, average price, sentiment score, and correlation coefficient of 0.52. Additionally, filters such as property type, room type, and price are also included in the dashboard.

Visualization 1 - Sentiment-Price Correlation: Grouped by AFINN categories, Very Positive listings (\$143) demonstrate 18% premium over Neutral (\$121). Hosts can improve pricing through positive, trust-building language emphasizing comfort and experiences. Hosts can improve pricing power by using more positive, vivid, and trust-building language in listing descriptions

Visualization 2 - Topic Performance: The visualization is composed of six topics and is positioned based on price, reviews, and sentiment. It is identified that beach-themed properties perform 35% better than urban properties at \$115. Hosts should build review momentum early while preserving distinctive value in their listing experience and description.

Visualization 3 - Market Competitiveness: The visualization is composed of various groups of reviews and is used to identify the optimal number of reviews for credibility and pricing power at 50-100 reviews and a price of \$138. Additionally, new properties with 0-10 reviews average at \$118, while properties with 500+ reviews experience a commoditization of their properties at \$125. Hosts should build review momentum early while preserving distinctive value in their listing experience and description.

Visualization 4 - Property Pricing: Property room configurations show certain combinations (Entire Home: \$165; Shared Rooms: \$95) create 3.5x variance. Placing within proper segments according to amenities is far more important than general pricing competition. Airbnb can segment hosts more effectively by property and room configuration, because these combinations show clear differences in value perception and pricing.

Key Findings and Strategic Implications

Airbnb listings show extremely positive sentiment with dominant emotions of trust, joy, and anticipation. The correlation between positive sentiment and pricing ($r = 0.52$) supports the quality of description impacting perceived value. TF-IDF analysis indicates 70% of descriptions contain generic terms while descriptions of unique terms correlate to 30% higher engagement. Topic modeling indicates convenience of location, privacy, and amenities differentiate properties; beach themes command 20-30% premium pricing. Review Volume analysis indicates 50-100 reviews is optimal for credibility and pricing power.

Strategic recommendations include utilizing sentiment scoring for new listings, creating templates for topic-based content focusing on unique features, and implementing review velocity campaigns targeting 50-100 reviews. Sentiment scores can be used to command 15-25% pricing premiums for high-quality content. Overall revenue potential is projected to be 30-40% through multiplicative effects of sentiment improvement, topic modeling, and positioning optimization. Listing descriptions are used as strategic tools to impact pricing and performance beyond their role as descriptions.